Q1. Database Technologies

Question 1.1

What does CRUD stand for and what are the four fundamental database operations?

Answer:

CRUD stands for Create, Read, Update, Delete. These are the four fundamental database operations that work with any database system.

Question 1.2

In the three-layer data model architecture, which formats are used for each layer? (Hint: the formats are Dart objects, Database records, and JSON)

Layer	Format	Purpose
Dart Class	?	Type Safety & Code Clarity
Database	?	Persistence & Reliability
API Communication	?	Portability & Interoperability

Answer:

Layer	Format	Purpose
Dart Class	Dart objects	Type Safety & Code Clarity
Database	Database records	Persistence & Reliability
API Communication	JSON	Portability & Interoperability

Question 1.3

- **??1**: The data itself (like books in a library)
- **??2**: software that manages data (like the librarian who organizes and helps you find books)
- **??3**: The computer system hosting the ??1 and ??2 (like the library building that houses both books and librarian)

Which one is Database, DBMS, and DB Server?

Answer:

- **Database**: The data itself (like books in a library)
- **DBMS**: software that manages data (like the librarian who organizes and helps you find books)
- **DB Server**: The computer system hosting the DBMS and database (like the library building that houses both books and librarian)

Question 1.4

Complete the following SQL vs NoSQL comparison table:

SQL Concept	NoSQL Equivalent
Database	?
Table	?
Row	?
Column	?

Answer:

SQL Concept	NoSQL Equivalent
Database	Database (folder of collections)
Table	Collection (folder of documents)
Row	Document (JSON object)
Column	Field (key-value pair in JSON)

Question 1.5

Write equivalent CRUD operations for both SQL and MongoDB NoSQL to:

1. Create a student record with id=3, name="Charlie", age=21
2. Read all students
3. Update student with id=3 to age=22
4. Delete student with id=3

Answer:

SQL:

```sql
-- Create
INSERT INTO Students (id, name, age) VALUES (3, 'Charlie', 21);

-- Read
SELECT * FROM Students;

-- Update
UPDATE Students SET age = 22 WHERE id = 3;

-- Delete
DELETE FROM Students WHERE id = 3;
```

MongoDB NoSQL:

```javascript
// Create
db.Students.insertOne({id: 3, name: "Charlie", age: 21});
```

Question 1.6

Classify the four databases used in ASE 456 course:

Database	Type (SQL/NoSQL/Hybrid)	DB/DB Server
PocketBase	?	?
IndexedDB	?	?
SQLite	?	?
Firebase	?	?

Answer:

Database	Type (SQL/NoSQL/Hybrid)	DB/DB Server
PocketBase	Hybrid	DB Server
IndexedDB	NoSQL	DB
SQLite	SQL	DB
Firebase	NoSQL	DB Server

Question 1.7

Given this Dart class, show how the same data would be represented in:

1. Dart object
2. SQL INSERT statement
3. JSON format

```
class Todo {
  final int id;
  final String title;
  final bool completed;
}
```

Example data: id=1, title="Learn Flutter", completed=false

Answer:

1. Dart Object:

```
Todo todo = Todo(
  id: 1,
  title: "Learn Flutter",
  completed: false
);
```

2. SQL INSERT:

```
INSERT INTO todos (id, title, completed)
VALUES (1, 'Learn Flutter', false);
```

3. JSON:

```
{
  "id": 1,
  "title": "Learn Flutter",
```

Question 1.8

(T/F): In Dart, accessing a Database vs Database Server requires different code and APIs.

Explain your answer.

Answer:

False

In Dart, accessing Database vs Database Server uses the same API/driver approach, so the CRUD code looks identical. The difference is in deployment and management:

- **Database**: The data storage itself
- **Database Server**: Software that hosts databases and handles security, concurrency, and transactions

From a Dart developer's perspective, both are accessed through APIs, making the usage feel the same.

Question 1.9

(T/F) SQL databases use specialized data structures:

- **Tables**: Stored as files on disk with rows and columns
- **Indexes**: Often implemented as B-Trees for fast search operations
- **Relations**: Connections between tables using foreign keys
- **Transactions**: Ensure ACID properties (Atomicity, Consistency, Isolation, Durability)

Answer:

T

Summary

We have 9 questions on this page.

1. How many questions did you answer? (/ 9)
2. What percentage of questions did you answer? (%)
3. List question numbers that you cannot answer:
4. (Optional) Explain why you could not answer these questions.